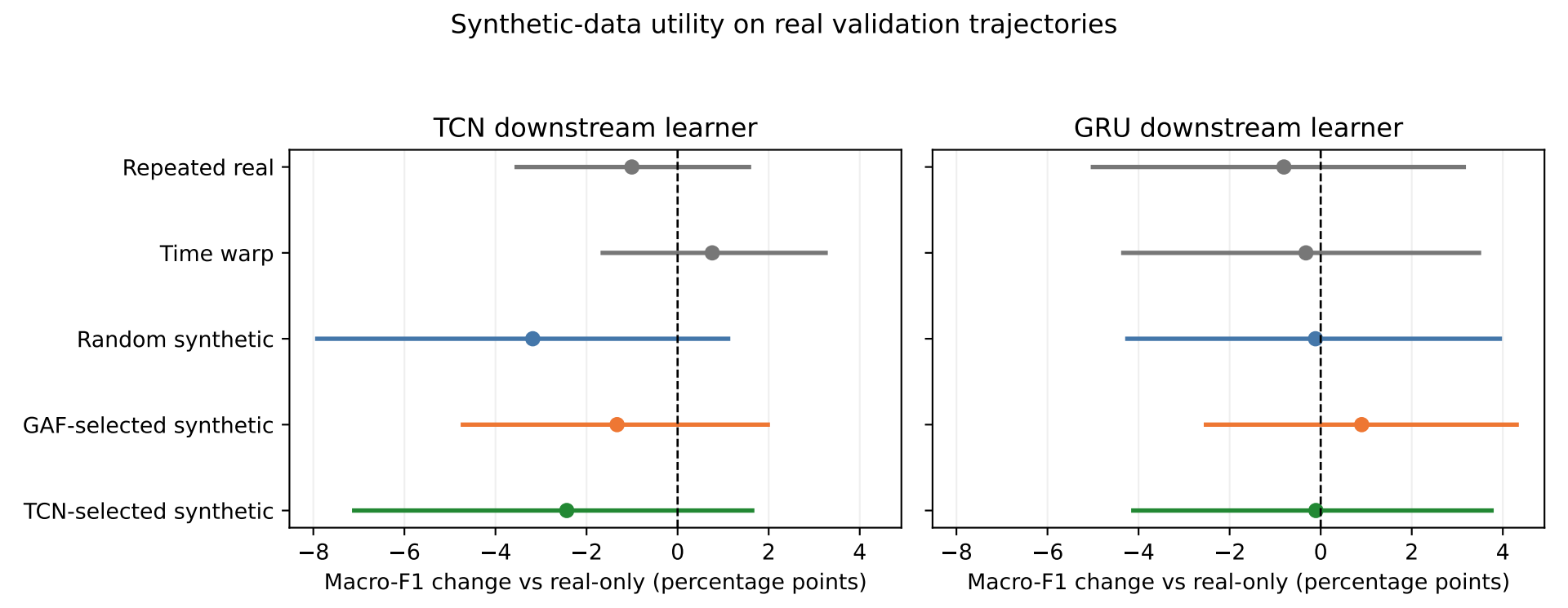
Synthetic-data utility on real validation trajectories
TCN downstream learner
GRU downstream learner
Repeated real
Time warp
Random synthetic
GAF-selected synthetic
TCN-selected synthetic
−8
−6
−4
−2
0
2
4
Macro-F1 change vs real-only (percentage points)
Macro-F1 change vs real-only (percentage points)
Paired 95% vehicle-bootstrap intervals; three learner seeds, one generator. Exploratory validation, not final test.